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A promising Co-P/Ni-P multi-layer barrier layer for Bi_2Te_3 based thermoelectric device

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ABSTRACT

Ni layer and Co layer are the most commonly utilized in commercial Bi_2Te_3-based thermoelectric device as diffusion barrier layer. However, during the application they still have significant limitations in inhabiting the diffusion of elements. In this study, we found that the Ni layer exhibits a greater tendency to react with Te in Bi_2Te_3. In contrast, while Co is less prone to react with Bi_2Te_3, it facilitates interdiffusion with Sn in Sn-based solder at the interface. Therefore, a barrier layer based on p-type Bi_2Te_3 wafers with electroless plating Co-P (EC) and electroless plating Ni-P (EN) was investigated to avoid the drawbacks of the single barrier layer. Co-P/Ni-P barrier layer can be obtained by electroless plating Co and then Ni (ECEN). The Co-P/Ni-P barrier layer demonstrates an excellent diffusion blocking effect, effectively inhibiting the formation of the Ni-Te phase in the EN-plated samples and reducing diffusion between the barrier layer and Sn95Sb5 solder in both EC-plated and EN-plated samples. The intermetallic compounds (IMCs) at the interface of ECEN-plated samples were thinner after annealing compared to those at the interfaces of in the conventional EN-plated and EC-plated samples. The wettability of the ECEN-plated *p*-type Bi_2Te_3 surface exceeded that of both the EC-plated and bare surfaces. Ultimately, the bonding strength between p-type Bi_2Te_3 and the Co-P/Ni-P barrier layer was 10.12 MPa, and the interfacial contact resistance at room temperature was 3.34 $\mu\Omega\cdot cm^2$, meeting the requirements for the thermoelectric device fabrication and operation.

1. Introduction

In recent years, the increasing demands for sustainable energy solutions to prevent environmental degradation has highlighted the importance of thermoelectric materials, which offer significant advantages in terms of efficiency and cleanliness [1,2]. As the core component of the thermoelectric module, thermoelectric materials are used according to the Seebeck effect and the Peltier effect to achieve thermoelectric conversion. A wide variety of thermoelectric materials have been used in different operating temperature ranges, such as Bi_2Te_3 [3–5], PbTe [6,7], SnTe [8,9], and so on. Among many thermoelectric materials, Bi_2Te_3 has excellent thermoelectric properties near room temperature and have been widely used for commercial production [10,11].

The common Bi_2Te_3-based thermoelectric device comprises several components: thermoelectric materials, electrodes, interconnecting conductive sheets, encapsulation, and fillers. Thermoelectric materials form thermocouples that are connected in thermal parallelism and electrical series. These thermocouples are sandwiched between ceramic sheets to assemble the complete thermoelectric device [12,13]. However, the development of thermoelectric devices has significantly lagged behind that of thermoelectric materials. While thermoelectric materials can be expressed in terms of thermoelectric figure of merit (*ZT*) for thermoelectric properties, thermoelectric devices require an integrated approach after considering numerous factors. In addition to the thermoelectric materials, the interfacial contact between these components greatly affects the service characteristics of the thermoelectric devices, such as energy conversion efficiency, reliability, and fatigue life [14,15].

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Bi_2Te_3-based thermoelectric materials already possess advanced fabrication methods, including melt spinning [16], cold sintering [17], and 3D printing [18–20]. In the manufacturing of thermoelectric devices, welding is often used to connect thermoelectric materials with electrodes by Sn-based solder which is the most commonly used [21,22]. The reaction between Bi_2Te_3-based thermoelectric materials and Cu based electrodes, as well as with Sn-based solder, leads to the formation of brittle Sn-Te intermetallic compounds at the interface [23]. These compounds contribute to the sprouting and expansion of internal voids and cracks, thereby affecting the interface properties, including bonding strength, contact resistance, and thermal stability [24–26]. To ensure the reliability of the interfacial connection, a metal barrier layer is typically applied before welding the thermoelectric material to the electrode. This involves plating a metal layer between the thermoelectric material and the solder, as well as between the solder and the metal electrode. Common methods for preparing barrier layers include electroless plating, electroplating, and spraying [27]. Given the additive manufacturing processes of inorganic materials, advanced 3D printing technologies are highly likely to be feasible for preparing barrier layers. Among the metals used for barrier layers, nickel is more widely studied and commonly used [28,29]. The presence of Ni makes a dense layer of Ni-Te phase generated at the interface near the solder side, which effectively avoids the reaction between the metal electrode material, Sn-based solder and Bi_2Te_3-based thermoelectric material to some extent [30,31], and reduces the rate of internal resistance increase [32]. However, most of the Ni plating is currently carried out by magnetron sputtering, electrochemical deposition, and thermal spraying [33], resulting in the lower bond strength and poor stability. And the increase in the thickness of the Ni-Te layer during working leads to the loss of Te in the thermoelectric material, which deteriorated the thermoelectric properties of the whole thermoelectric module [34].

Considering the drawbacks of Ni as a barrier layer, Co-coated thermoelectric materials have higher Seebeck coefficients than Ni-coated thermoelectric materials [35], and the coefficients of thermal expansion of metal Co and Bi_2Te_3-based thermoelectric material are similar, so Co has been investigated as a promising barrier layer [36]. Wang et al. [37,38] investigated the compositional changes of intermetallic compounds (IMCs) between Co and Sn-based solder at phosphorus (P) contents of 0.7 wt% and 6 wt%. They found that the growth of $CoSn_3$ was inhibited in the specimen with the addition of 0.7 wt% P. Furthermore, the Co-P layer with the lower P content exhibited excellent diffusion blocking properties. Li et al. [39] plated a 6.2 ± 0.5 μm thick Co-P layer between In-48Sn solder and Bi_2Te_3. After aging treatment, no significant diffusion was observed between the Co-P layer and Bi_2Te_3. Although diffusion occurred in both Co and In-48Sn solder, leading to the formation of different intermetallic compounds (IMCs), the growth of these IMCs was slow. It was concluded that the Co-P layer is an effective diffusion barrier for both In and Sn. Wang et al. [40,41] found that the presence of Co-P layer can inhibit the generation of brittle IMCs such as SnTe, resulting in an increase in the shear strength of *n*-type and *p*-type samples by a factor of 1.5 and 7, respectively, with no significant degradation in the thermoelectric properties. These results suggest that Co may be suitable to replace Ni as an excellent barrier layer.

However, we found that the Co-P barrier layer still has its drawbacks. For example, The Co-P layer is prone to oxidation in air. It is conducive to analysis how to enhance the high-temperature stability and mechanical reliability of thermoelectric modules by comparing the interfaces between the Co-P barrier layer and Sn95Sb5 solder with that between the Ni-P barrier layer and the same solder. Suppressing the interfacial reactions between the barrier layer and the solder is crucial for ensuring the reliability and lifespan of thermoelectric modules. Besides, the optimization of both Ni-P and Co-P layer is mostly limited to single-layer process optimization in the existing work reported.

In this study, we investigated the performances of the interfaces between the Co-P barrier layer and Sn95Sb5 solder and that between the Ni-P barrier layer and the same solder, respectively. Ultimately, a new surface modification process for *p*-type Bi_2Te_3 was developed by electroless plating Co-P/Ni-P. The interfacial reaction of *p*-type Bi_2Te_3/diffusion layer/solder was analyzed and the wettability of Sn95Sb5 solder on Co-P layer was measured. Moreover, the bonding strength and contact resistivity of thermoelectrical modules with Co-P/Ni-P layer were determined and compared to that with Ni-P layer and Co-P layer. It is demonstrated that, compared to individual Ni-P and Co-P barrier layers, the Co-P/Ni-P barrier layer could be a superior diffusion barrier for *p*-type Bi_2Te_3 and Sn95Sb5 solder.

2. Experimental

The Bi_2Te_3-based thermoelectric wafers used in this study were all commercially available and Sb-doped to obtain *p*-type Bi_2Te_3 with a composition of $Bi_{0.4}Sb_{1.6}Te_3$. Before electroless plating, the *p*-type Bi_2Te_3 wafers were ultrasonic cleaned with acetone to remove the oil and organic impurities on the surface of the wafers, and the *p*-type Bi_2Te_3 wafers were ground and polished with 0.1μm Al_2O_3 suspensions. The *p*-type Bi_2Te_3 wafers were then submerged in a mixed solution of HNO_3 and HCl for etching to remove inorganic impurities and increase the surface roughness, and finally the wafers were cleaned with deionized water. Afterward, the wafers were sensitized, cleaned with weakly alkaline water and then activated to ensure the formation of a catalytic film on the wafer surface and the generation of a sufficient number of nuclei to facilitate the deposition reaction in the electroless plating process. The surfaces of *p*-type Bi_2Te_3 wafers were electroless plated with Ni-P barrier layer, Co-P barrier layer, and Co-P/Ni-P composite plating respectively at a temperature around 80 °C ~ 90 °C. The schematic diagram of the electroless plating Co-P/Ni-P process is shown in Fig. 1.

Sn95Sb5 solder was soldered onto the surface of wafers with different barrier layers by reflow soldering, and then the wafers were placed in a muffle furnace for solid-state aging treatment at an annealing temperature of 180°C and annealing time of 24 h, after which the wafers were cooled to room temperature in air. The microstructures were studied by field scanning electron microscopy (FESEM) (FEI, Nova 400 Nano SEM) and the elemental compositions were determined using energy-dispersive X-ray spectroscopy (EDS). The phase identification was characterized by X-ray diffraction with an X'per Pro diffractometer using Cu Kα radiation (λ=1.5418 Å). Solder spreadability was calculated as the percentage decrease in height of 0.76 mm solder ball that was placed on the plated wafers during heating at 270℃ for 10 min. Contact angles were evaluated from the cross-section view of solder balls on the plated *p*-type Bi_2Te_3 wafers surfaces. The bonding strength measurement was performed by cutting the plated *p*-type Bi_2Te_3 wafer into 3 × 3 × 1 mm, and then welding the tensile sample with two copper blocks to form a sandwich structure with Sn95Sn5 solder, which helps to apply a tensile load. The contact resistivity was obtained by the four-probe method via the conductivity-Seebeck coefficient scanning probe microscope.

3. Results and discussion

3.1. Interfacial reactions of p-type Bi_2Te_3/barrier layer/solder

Fig. 2 shows the cross-sectional SEM image and the backscattered images of *p*-type Bi_2Te_3/Ni-P/solder after annealing for 6 h and 24 h at 180°C. Concentrated Ni-P layer was found between *p*-type Bi_2Te_3 and Sn9Sb5 solder. Sn from the solder (the right of Ni-P layer) was not detected on the left of the Ni-P layer, thus demonstrating that Ni-P layers effectively inhibit the diffusion of Sn [42]. After annealing for 24 h, Fig. 2(e) shows the EDS element line profiles and Fig. 2(f) shows the corresponding EDS results of the point 1, 2. 3. 4 and 5. Contrary to the results obtained after 6 h of annealing, the EDS element line profile of Ni in Fig. 2(e) revealed a small Ni peak on the right side of the Ni-P layer, for example, the proportion of Ni at point 4 is reaches as high as 34.30 at

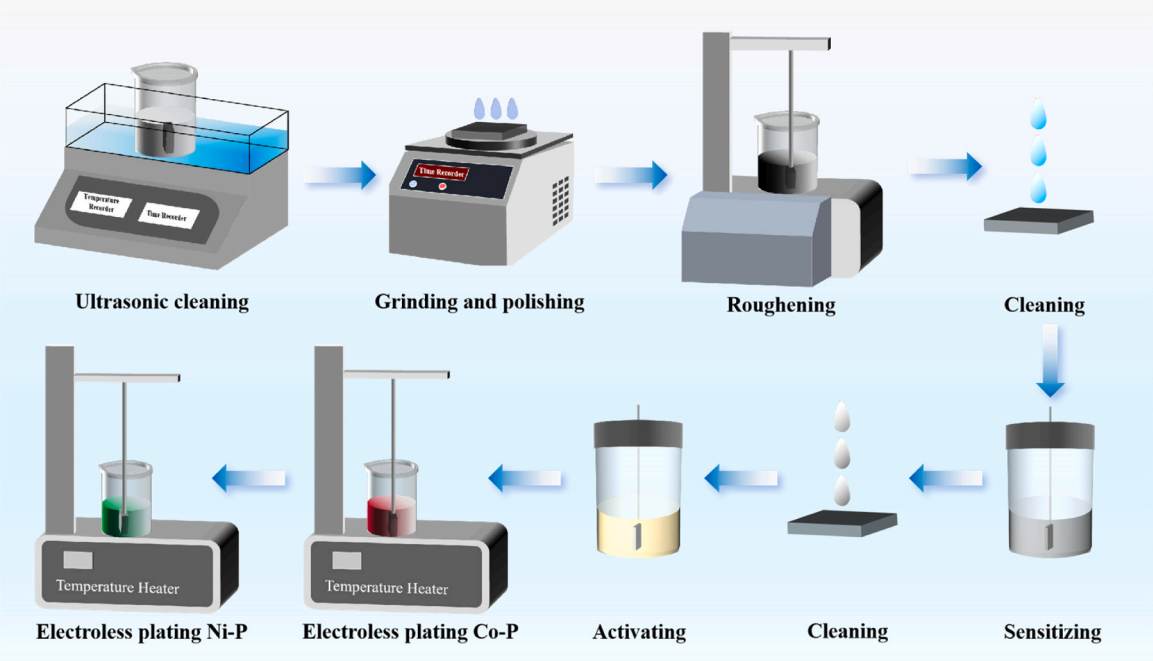


Fig. 1. Schematic diagram of the electroless plating Co-P/Ni-P process.

%. It indicates that Ni-Sn IMCs may form between Ni-P and solder, and the IMC is highly probable to be Ni_3Sn_4. Because judging from the phase diagram, a total of three phases, Ni_3Sn, Ni_3Sn_2, and Ni_3Sn_4, are theoretically generated at low temperatures [43,44]. But the Ni_3Sn_4 phase or sub-stable phase is more easily generated than the other two phases [45], and only Ni_3Sn_4 of the three phases grows stably in the solid-phase reaction annealed at 180 °C. In addition, a small amount of Ni was detected in the *p*-type Bi_2Te_3 on the left side of the Ni-P layer in the sample after 24 h of annealing, for example, the proportion of Ni at point 2 is about 1.17 at%. It indicats that Ni has diffused into the thermoelectric material as well.

Fig. 3 illustrates the cross-sectional SEM image, the backscattered images and the EDS results of *p*-type Bi_2Te_3/Co-P/solder after annealing for 6 h and 24 h at 180°C. As shown in Fig. 3(a) and (d), Sn from Sn95Sb5 solder was not detected on the left of Co-P layer even after annealing 24 h. This indicates that under this annealing condition, the Co-P layer can also prevent Sn from diffusing into the *p*-type Bi_2Te_3. But the mutual diffusion between the Co-P layer and Sn95Sb5 solder occurs more rapidly compared with that between the Ni-P layer and Sn95Sb5 solder on the contrary, leading to a significant fading of the Co-P layer. It can be seen in Fig. 3(d) that the Co-P layer between *p*-type Bi_2Te_3 and the solder became thin and uneven, and the gray clusters formed between Co-P layer and solder. It also can be seen from the EDS results in Fig. 3(e) that there were no Co detected at point 1 and 2, and the composition of point 4 at the grey cluster is Co 76.35 at%, P 11.49 at%, Sn 12.16 at%. This result is considered by other researchers to indicate that different IMCs (a mixture of multiphase compounds of Co-Sn-P) may have been formed at point 4 [46].

The main reasons for the differences between the reactions occurring in barrier layers and solder may lie in the distinct crystal structures and surface structures of the barrier layers [47]. As the results of GIXRD shown in Fig. 4(a), the Co-P phase contains both crystalline and amorphous structures, while the Ni-P phase is amorphous. Compared with amorphous structures, crystalline structures have grain boundaries and more rapid diffusion paths [48,49]. For the Co-P layer with mixed microstructure, both Sn and Co atoms will diffuse. The Sn diffuses more easily into the mixed structural Co-P layer, Co diffuses toward the solder to form Sn-Co IMC and the diffusion of Co happens non-uniformly [50]. Meanwhile, as shown in Fig. 4(b) and (c), the microscopic surface particles of Co-P are tightly connected and smaller in size, while those of Ni-P are larger and have a "cellular" structures. Smaller particle sizes imply a higher number of particles per unit area, and consequently, these particles provide a greater number of surface active sites. Active sites can act as initiation points for chemical reactions, thereby increasing the likelihood of interaction with the solder [51].

The orientation degree *F* for specific crystal planes along the out - of - plane direction of Co-P layer were calculated using the Lotgering method:

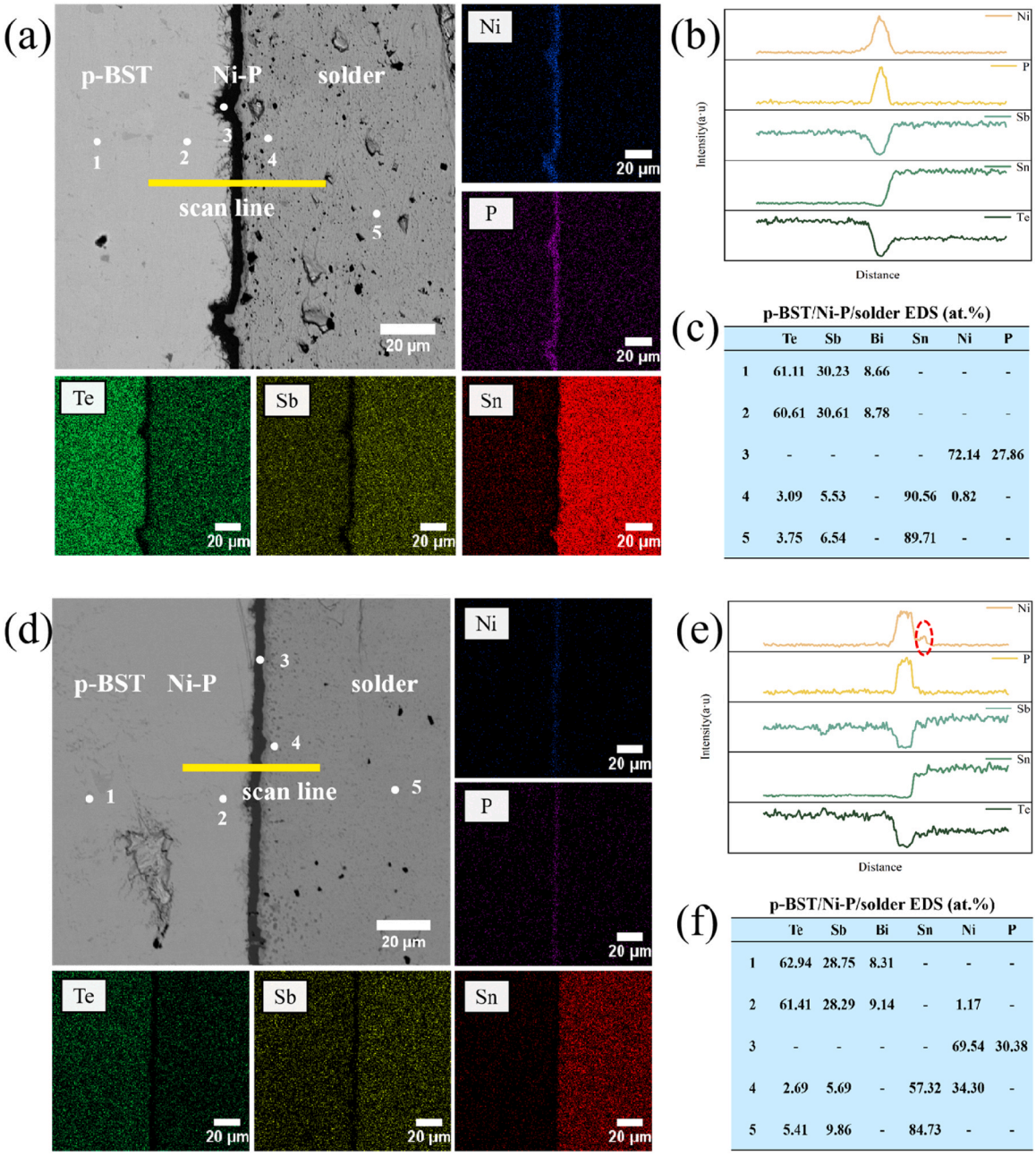
$$F = (P - P_0)/(1 - P_0) \tag{1}$$

$$P_0 = I_0(h\quad k\quad l)\Big/\sum I_0(h\quad k\quad l) \tag{2}$$

$$P = I(h\quad k\quad l)\Big/\sum I(h\quad k\quad l) \tag{3}$$

Here, I and I_0 represent the peak integral intensities of the measured sample and a fully randomly oriented sample, respectively. The calculated orientation degree of (0 0 2) plane ($F_{(0\ 0\ 2)} = 0.14$) is much lower than that of (1 0 0) plane ($F_{(1\ 0\ 0)} = 0.28$) and (1 0 1) plane ($F_{(1\ 0\ 1)} = 0.26$). These values suggest the (0 0 2) plane exhibits a preferred orientation along the in-plane direction of the Co-P layer. Further confirmation can be obtained through pole figure and inverse pole figure analysis. The XRD pole figure and inverse pole figure results for Co-P are shown in Fig. 5. The relatively high-intensity regions in (0 0 0 2) pole figure and Z inverse pole figure also indicate the preferred orientation of the (0 0 0 2) plane along the in-plane direction of the Co-P layer, consistent with the result calculated by Lotgering method. Ni-P, being amorphous and lacking a periodic structure, cannot form a regular orientation distribution. The preferred orientation of the (0 0 0 2) plane along the in-plane would cause diffusion to preferentially occur along the (0 0 0 1) interfaces or crystallographic directions. Furthermore, the (0 0 0 1) interplanar spacing ($d_{(0\ 0\ 0\ 1)} = 4.000$ Å) is the largest among all other planes for Co-P layer. Therefore, in the Co-P layer, the presence of the preferred orientation of the (0 0 0 2) plane along the in-plane enhances the directed diffusion rate of Co, accelerating the consumption of Co-P.

Fig. 6 illustrates the cross-sectional SEM image and EDS results of the *p*-type Bi_2Te_3/Co-P/Ni-P/solder. As shown in Fig. 6(a) an (d), the Co-P/



	Te	Sb	Bi	Sn	Ni	P
1	61.11	30.23	8.66	-	-	-
2	60.61	30.61	8.78	-	-	-
3	-	-	-	-	72.14	27.86
4	3.09	5.53	-	90.56	0.82	-
5	3.75	6.54	-	89.71	-	-

p-BST/Ni-P/solder EDS (at.%)

	Te	Sb	Bi	Sn	Ni	P
1	62.94	28.75	8.31	-	-	-
2	61.41	28.29	9.14	-	1.17	-
3	-	-	-	-	69.54	30.38
4	2.69	5.69	-	57.32	34.30	-
5	5.41	9.86	-	84.73	-	-

Fig. 2. Cross-sectional SEM images and backscattered images of p-type Bi_2Te_3/Ni-P/solder: (a) annealing for 6 h; (d) annealing for 24 h; the corresponding EDS element line profiles: (b) annealing for 6 h; (e) annealing for 24 h; the corresponding EDS result of the points: (c) annealing for 6 h; (f) annealing for 24 h.

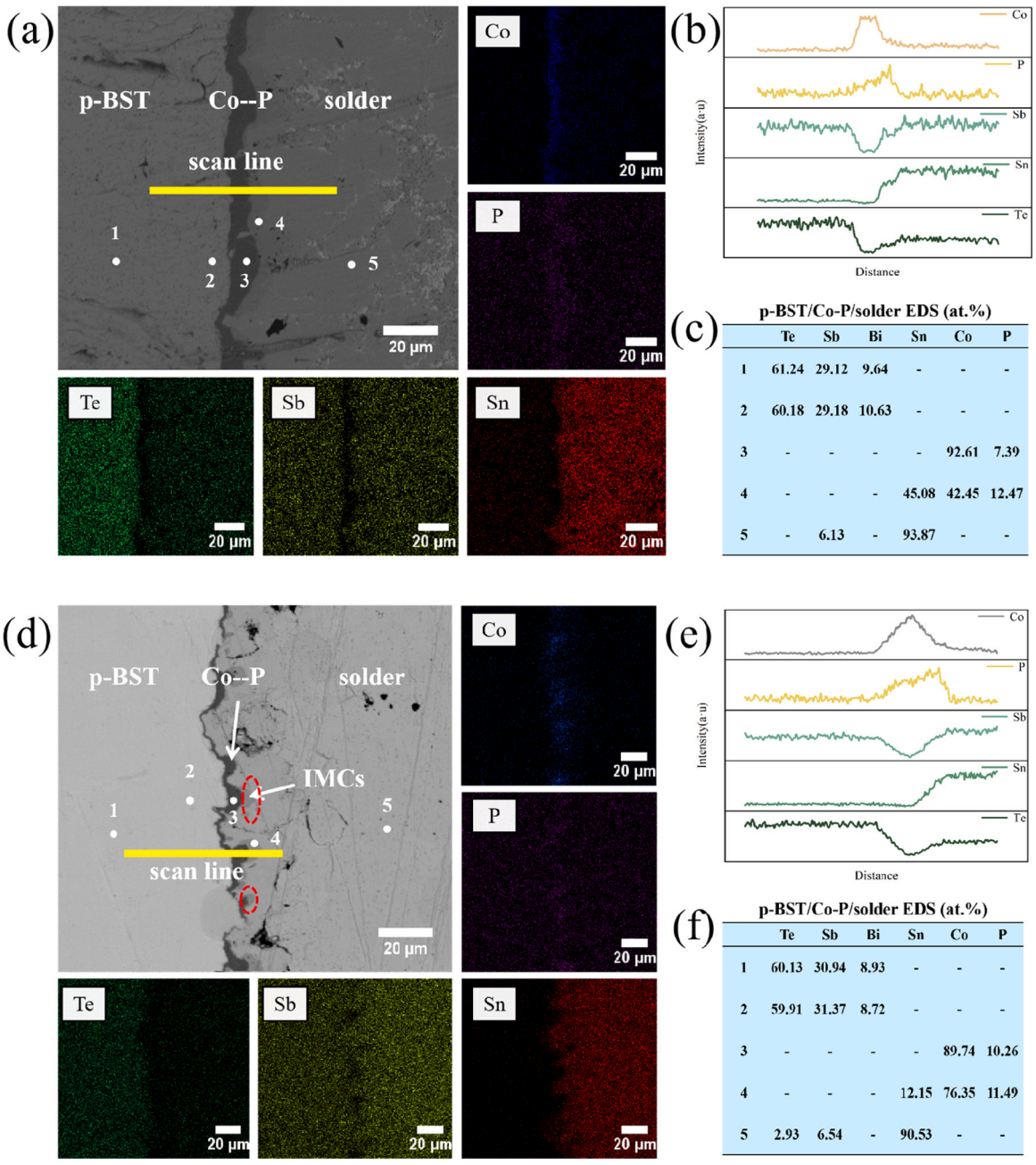
Ni-P layer, located at the interface between *p*-type Bi_2Te_3 and the solder, is enriched and effectively prevents the diffusion of Sn from the solder after annealing for 24 h. Additionally, it shows no significant formation of intermetallic compounds (IMCs), and the thickness of Co-P/Ni-P layer remains unchanged almost. As shown in Fig. 6(b) and (e), the EDS element lines profile of Sn confirmed the same conclusion. The EDS elements line profile of P increased two times from left to right, indicating that P content in Ni-P layer is higher than that in Co-P layer. The EDS element lines profile of Ni near the solder side did not show a small peak like that in Fig. 2(e), indicating that there no mutual diffusion occurred between Ni and Sn. The absence of Co and Ni in point 1, 2 of Fig. 5(c) and (f) suggested that the Co-P/Ni-P layer is effective in inhabiting the diffusion between *p*-type Bi_2Te_3 and barrier layer. In summary, the Co-P/Ni-P layer effectively blocks the diffusion of elements, outperforming both the Ni-P and Co-P layers.

3.2. *Wettability of Co-P diffusion barrier layer and Ni-P diffusion barrier layer*

The solder spreadability can be calculated through the change in the spread area:

$$SR = \frac{S_{final} - S_{initial}}{S_{initial}} \times 100\% \quad (4)$$

where S_R, S_{final}, and $S_{initial}$ are the spread rate, the area after solder spreading, the area before solder spreading. The average values of solder spreadability is displayed in Fig. 7. The pictures in Fig. 7 show a top-view of the spread solder ball on the surface of the sample after bonding. The *p*-type Bi_2Te_3 surfaces without plating, denoted as"No Plating", exhibited the poorest solder spreadability, with a result of 9.82 %. The solder spreadability of EN (52.8 %) was better than that



	Te	Sb	Bi	Sn	Co	P
1	61.24	29.12	9.64	-	-	-
2	60.18	29.18	10.63	-	-	-
3	-	-	-	-	92.61	7.39
4	-	-	-	45.08	42.45	12.47
5	-	6.13	-	93.87	-	-

p-BST/Co-P/solder EDS (at.%)

	Te	Sb	Bi	Sn	Co	P
1	60.13	30.94	8.93	-	-	-
2	59.91	31.37	8.72	-	-	-
3	-	-	-	-	89.74	10.26
4	-	-	-	12.15	76.35	11.49
5	2.93	6.54	-	90.53	-	-

Fig. 3. Cross-sectional SEM images and backscattered images of p-type Bi_2Te_3/Co-P/solder: (a) annealing for 6 h; (d) annealing for 24 h; the corresponding EDS element line profiles: (b) annealing for 6 h; (e) annealing for 24 h; the corresponding EDS result of the points: (c) annealing for 6 h; (f) annealing for 24 h.

with EC (20.48 %) and No Plating (9.82 %) surfaces, and it would greatly benefit the bonding.

Fig. 8 shows the average values of contact angle which correspond well with the result of the value of solder spreadability. The solder balls on both non-plated and EC-plated *p*-type Bi_2Te_3 surfaces retained their original spherical shapes, whereas those on the EN-plated *p*-type Bi_2Te_3 surfaces formed into hemisphere shapes. The contact angle of between the solder ball and the EN-plated *p*-type Bi_2Te_3 surfaces (92.79°) was the lowest. These results indicate that Ni-P surfaces exhibits the best wettability. It is probably due to the fact that the surface atoms of Ni-P amorphous structure are in an unsaturated-bond state with higher surface energy [52]. The difference in wettability of Sn95Sb5 solder on Co-P and Ni-P layers, which serve as substrates, is attributed to the distinct surface energies and crystallinities of Co-P and Ni-P layers. However, due to the unique atomic structure of amorphous alloys, there is currently no formula that can accurately calculate their surface energy. The characterization of wettability generally relies on Young equation:

$$\sigma_{sv} = \sigma_{sl} + \sigma_{lv}\cos\theta \tag{5}$$

where θ is the contact angel, σ_{ij} is the interfacial free energy and substrates *l*, *v*, and *s* represent the liquid, vapor, and solid states, respectively. In this study, the Ni-P substrate is amorphous, while the Co-P layer exhibits crystallinity. Structural relaxation and crystallization transformation in amorphous materials can increase the surface energy. Generally, the surface energy of amorphous materials is higher than that of crystalline materials, i.e., $(\sigma_{sv})_{amorphous} > (\sigma_{sv})_{crystalline}$. During the welding process, increased interfacial reactions enhance the bonding between solid and liquid atoms, leading to a reduction in the interfacial free energy [52]. This results in an increase in cos [[__]]#952;, a

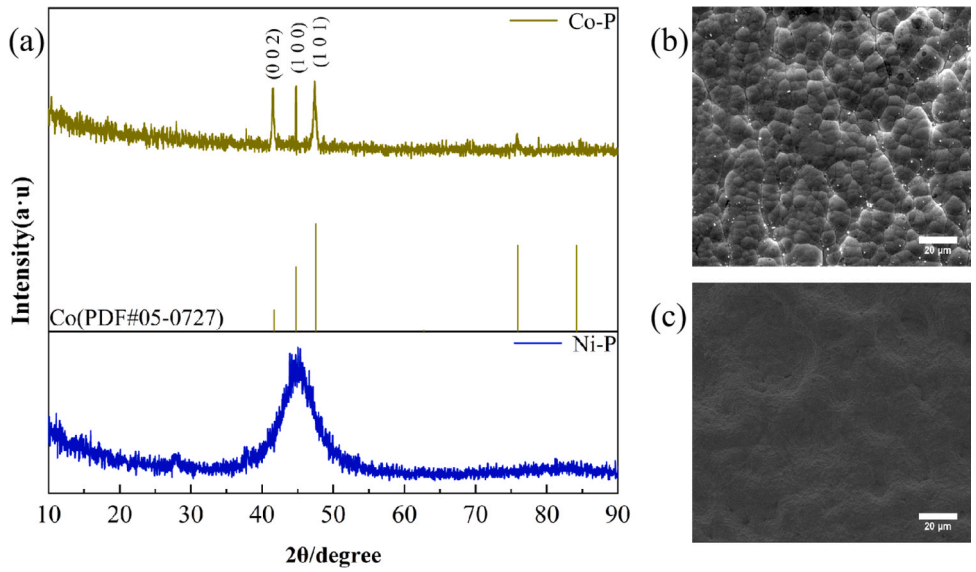


Fig. 4. (a) GIXRD results for Co-P and Ni-P; (b) surface morphology of electroless plating of Ni-P; (c) surface morphology of electroless plating of Co-P.

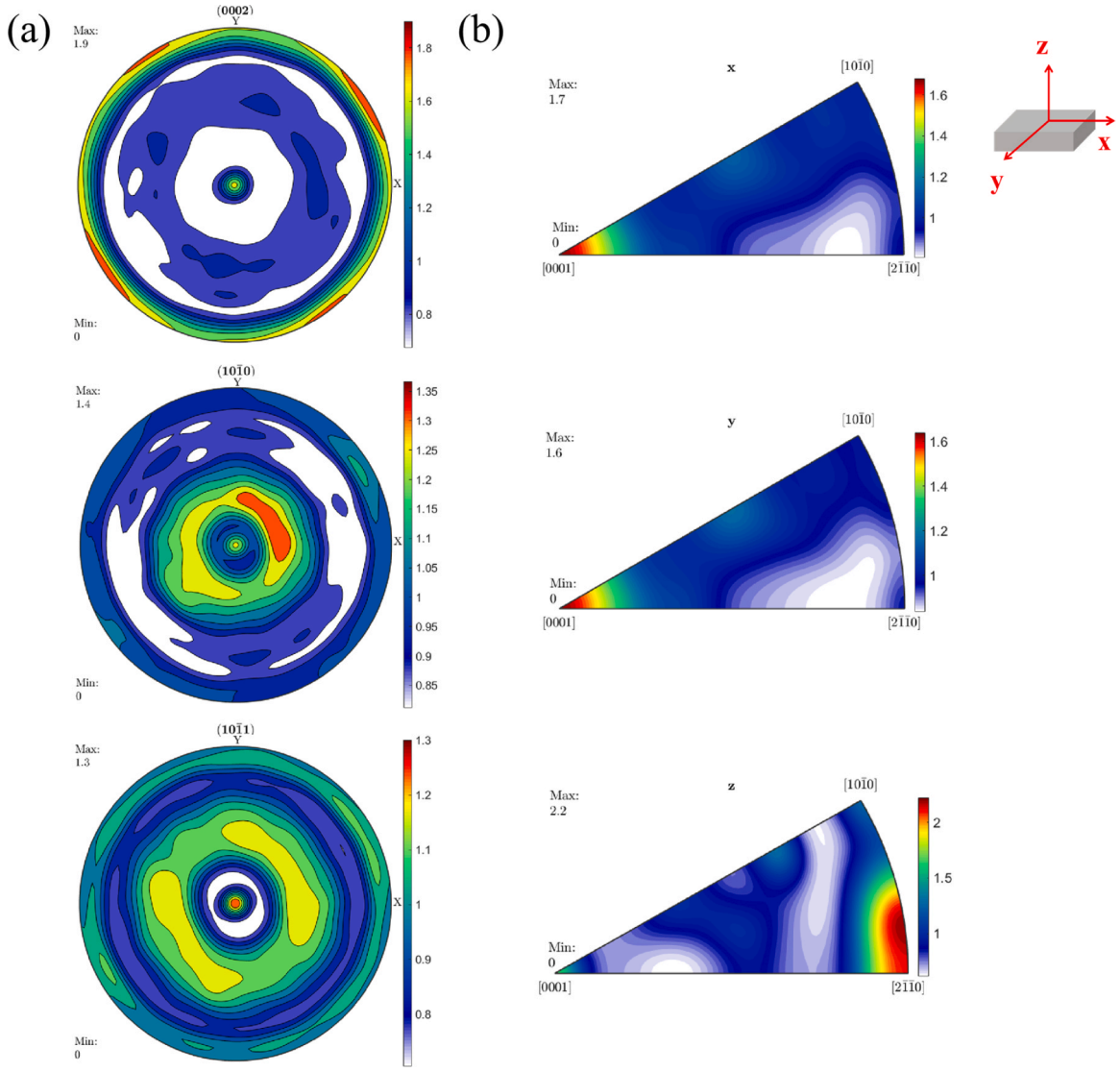
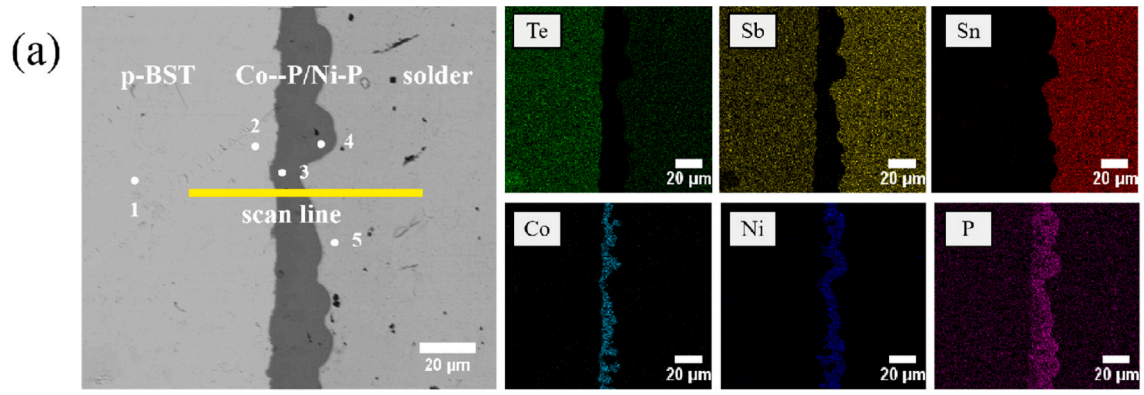
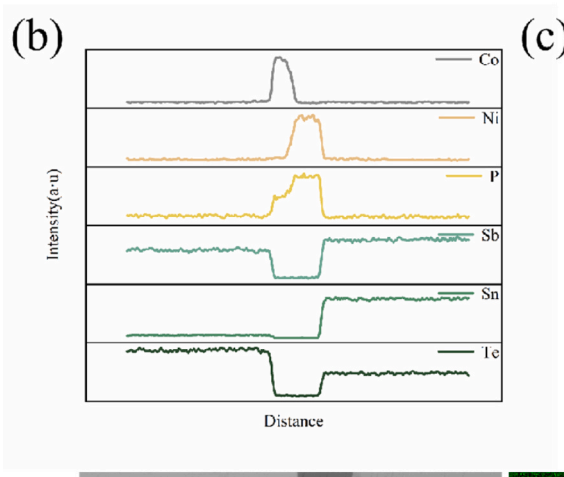


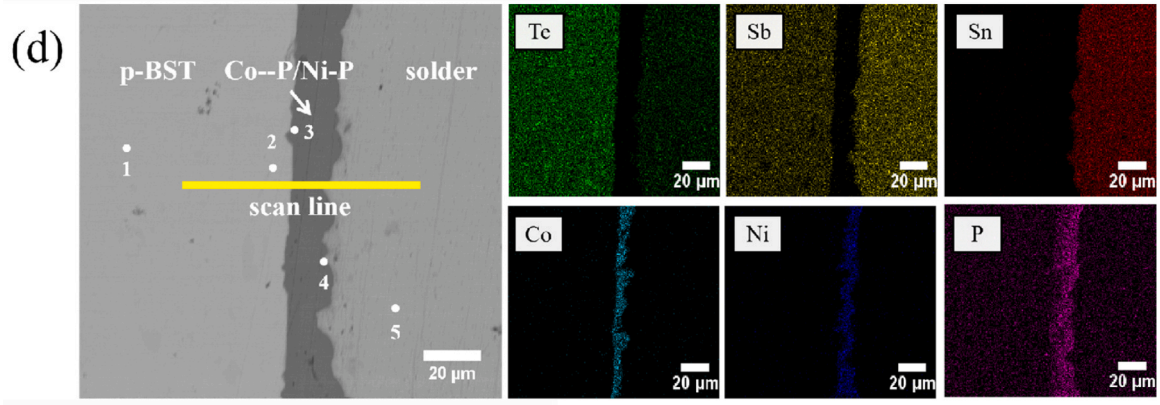
Fig. 5. (a) XRD pole figures of Co-P; (b)inverse pole figures of Co-P.

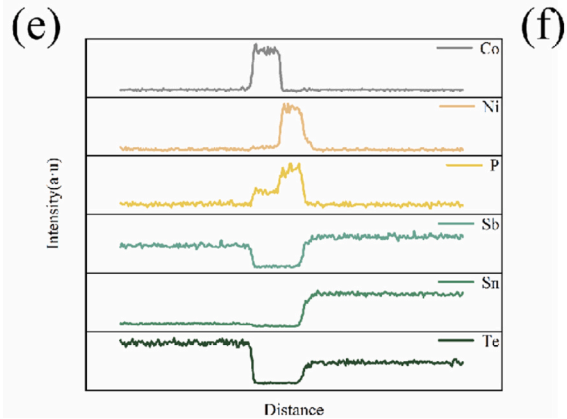




(c) p-BST/Co-P/Ni-P/solder EDS (at.%)

	Te	Sb	Bi	Sn	Co	Ni	P
1	63.17	28.88	7.95	-	-	-	-
2	61.19	29.89	8.91	-	-	-	-
3	-	-	-	-	85.68	-	14.32
4	-	-	-	-	-	79.68	20.32
5	3.97	8.44	-	87.59	-	-	-





(f) p-BST/Co-P/Ni-P/solder EDS (at.%)

	Te	Sb	Bi	Sn	Co	Ni	P
1	63.21	28.97	7.81	-	-	-	-
2	60.52	30.84	8.64	-	-	-	-
3	-	-	-	-	89.74	-	10.26
4	-	-	-	-	-	77.62	22.38
5	2.91	7.88	-	89.21	-	-	-

Fig. 6. Cross-sectional SEM images and backscattered images of p-type Bi_2Te_3/Co-P/Ni-P solder: (a) annealing for 6 h; (d) annealing for 24 h; the corresponding EDS element line profiles: (b) annealing for 6 h; (e) annealing for 24 h; the corresponding EDS result of the points: (c) annealing for 6 h; (f) annealing for 24 h.

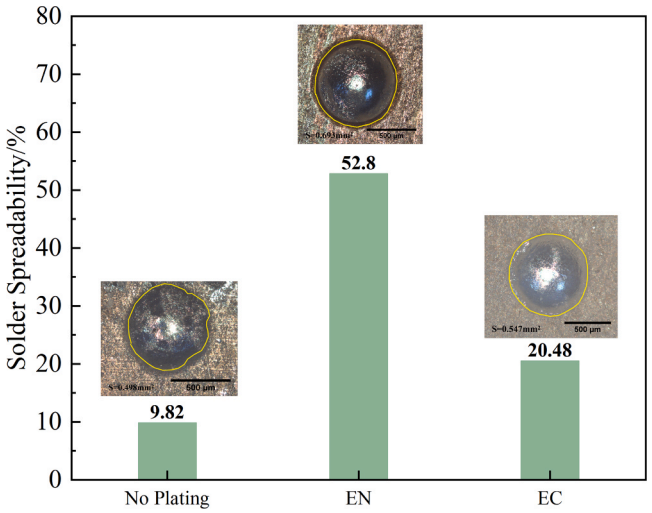


Fig. 7. Solder spreadability of Sn95Sb5 solder on Non-, EN-, and EC-plated p-Bi_2Te_3 surfaces.

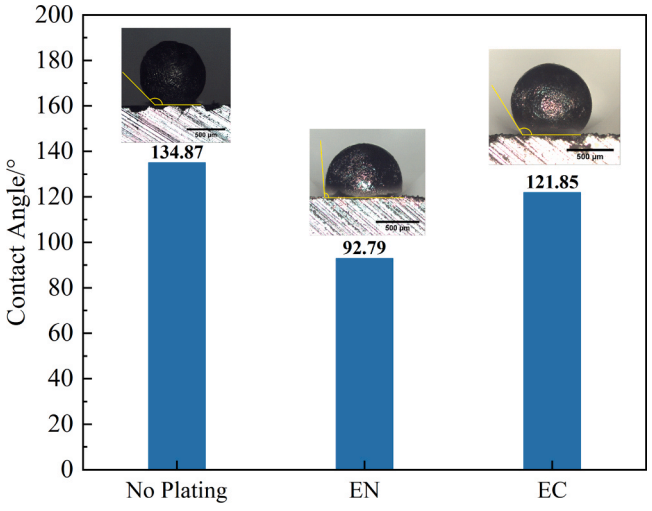


Fig. 8. Contact angle of Sn95Sb5 solder on Non-, EN-, and EC-plated p-Bi_2Te_3 surfaces.

decrease in θ, and an increase in the spreading rate (SR). Amorphous Co-based alloys tend to form dense oxide films, which is due to interfacial structural relaxation and primary crystallization reactions, leading to deteriorated wettability [53]. Although amorphous Ni-based alloys also form oxide films, the interfacial structural relaxation process leads to localized volume shrinkage, which allows liquid solder to penetrate between the alloy and the oxide film. This penetration accelerates the dissolution of the oxide film, and in turn, enhances wettability [54].

3.3. Bonding strength and contact resistance

Fig. 9 shows the bonding strength of EN-, EC-, ECEN-plated *p*-type Bi_2Te_3-based modules. The conventional EN, EC and developed ECEN-plated modules exhibited great bonding strength, with a bonding strength exceeding 10 MPa. The ECEN-plated thermoelectric modules, having the thickest barrier layer, experience increased stress, resulting in the lowest bond strength compared to the other modules. Nevertheless, this bond strength is still sufficient to satisfy the requirements for device fabrication. Fig. 10 (a)-(c) shows the contact resistivity results of different thermoelectric modules. The contact resistivity between a

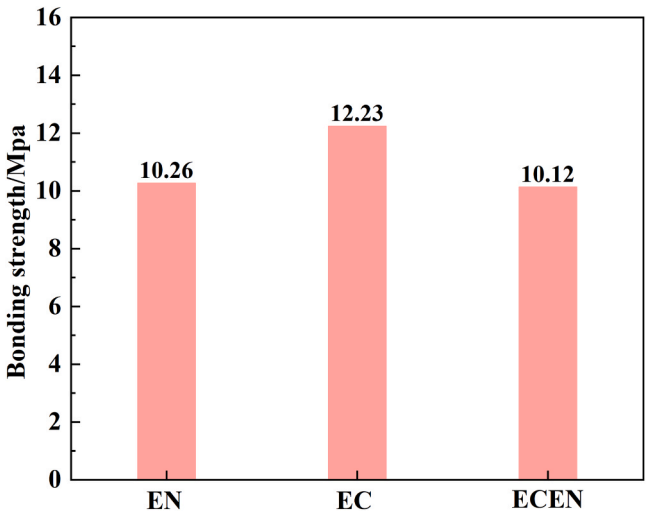


Fig. 9. (a) Bonding strength of p-type Bi_2Te_3-based thermoelectric modules with EN, EC, and ECEN barrier layers.

metal and a semiconductor is related to the Schottky barrier at the interface. According to metal-semiconductor contact theory, in the absence of the Fermi level pinning effect, the barrier height is equal to the difference in work function between the metal and the semiconductor. In existing research, it has been widely validated that measurements of current and voltage show no significant rectification characteristics for either forward or reverse currents. Therefore, the contact between p-BST and Ni/Co is assumed to be ohmic. When calculating the interfacial contact resistivity (R_C), it is necessary to consider the contact area between the two materials. The formula to calculate the contact resistivity is given by:

$$Rc = A\frac{\Delta V}{I} \tag{6}$$

Where Rc is the contact resistivity, A is the contact area of two materials, ΔV is the jumping voltage at the interface, I is the current at measurement. From the analysis of the curves in Fig. 10 (a)-(c), the contact resistivity of *p*-type Bi_2Te_3/Co-P/Ni-P/solder is 3.34 μΩ·cm^2 at room temperature. This value was in the middle range among the three modules and met the requirements for macro devices [55]. When considering only the interface contact resistivity R_c, the device figure of merit Z_d, and the material figure of merit Z_m, the following relationship is satisfied:

$$Z_d = Z_m\frac{1}{1+2R_c/\rho L} \tag{7}$$

Here, L is the length of the thermoelectric leg, and ρ is the density of the material. When L > 1 mm, the ideal value of R_c should be less than 10 μΩ·cm^2 [34]. In existing research, the interface contact resistivity for p-BST/Ni is 3.64 μΩ·cm^2, while the measured interface contact resistivity for p-BST/Co is below 3e-9 Ω·m^2 [41]. These results are consistent with the measurements obtained in this study. Generally speaking, contact resistivity depends on the thermoelectric material techniques, methods of forming their contacts with layers, processing methods such as pretreatment and cleaning of the thermoelectric material surface and some other factors [56]. In this study, the difference between EN-plated and EC-plated modules is attributed to the interfacial layer. The electrical conductivity of Co is better than Ni. Besides, Ni and Co are found to form ohmic contact, the interface formation energy for Co/Bi_2Te_3 interfaces is much lower than that of Ni/Bi_2Te_3 interfaces [57]. This suggests that the interface between Co and Bi_2Te_3 is more stable, and the band structure at the Co/Bi_2Te_3 interface have a lower potential barrier height, which is favorable for electron transport,

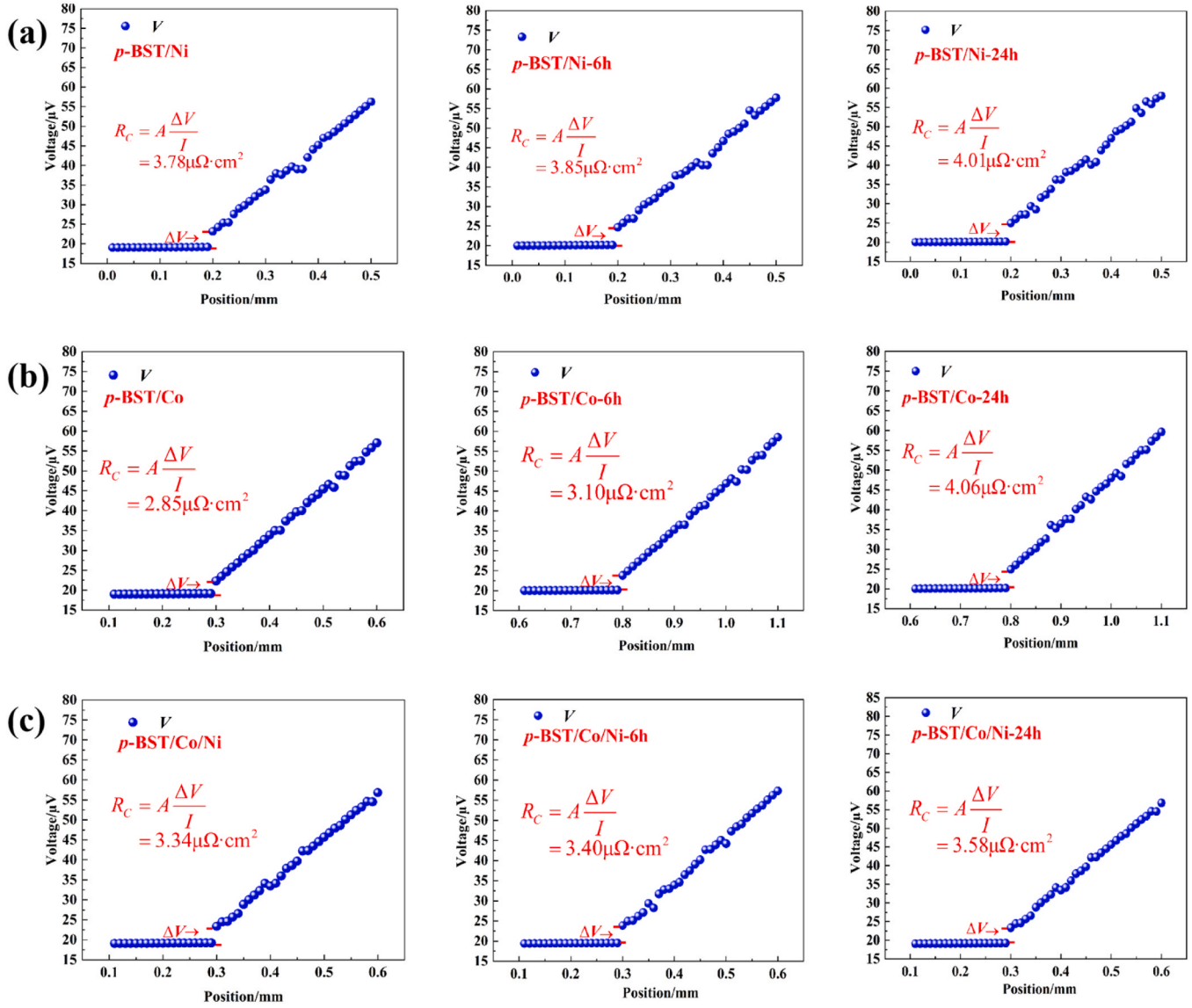


Fig. 10. (a)-(c) Contact resistivity of unannealed, 6h-annealed, 24h-annealed samples.

thereby resulting in a lower contact resistivity. After annealing at 180 ℃ for 6 and 24 h, the contact resistivity of the thermoelectric module with a Ni-P barrier layer gradually increased. After 24 h of annealing, the contact resistivity increased to 4.01 μΩ·cm^2. Similarly, after extending the annealing time, the contact resistivity of thermoelectric modules with a Co-P layer and a Co-P/Ni-P composite layer also gradually increased. However, there are differences: the contact resistivity of thermoelectric modules with a Co-P layer increased more significantly after 24 h of annealing, rising from 2.85 μΩ·cm^2 at room temperature to 4.06 μΩ·cm^2 after annealing, compared to those with Ni-P and Co-P/Ni-P layers. The reason for the inconsistent increase in contact resistivity with annealing time is that the Co-P layer is more prone to react with Sn to form brittle intermetallic compounds (IMCs). The growth of IMCs leads to an increase in contact resistivity. The contact resistivity of thermoelectric modules with a Ni-P barrier layer increases more significantly than that of modules with a Co-P/Ni-P composite layer. This is because the Ni-P layer is more prone to react with Te and Sb in p-BST compared to the Co-P layer. Thus, this is why the Co-P/Ni-P composite layer shows excellent performance with slower contact resistivity increase after annealing.

4. Conclusion

In this work, an electroless Co-P/Ni-P layer was selected as the diffusion barrier for the joints of *p*-type Bi_2Te_3 and Sn95Sb5 solder. Examination of the interfacial element distribution provided insights into the interfacial reactions during the annealing process. Upon annealing, the Ni-P layer reacts with *p*-type Bi_2Te_3. In comparison, the Co-P layer remains unreactive with *p*-type Bi_2Te_3. However, Co-P exhibits a faster reaction with Sn95Sn5, leading to significant consumption of the Co-P layer. Texture analysis of the barrier layer shows that the crystalline Co-P barrier layer has a preferred orientation, making it more reactive with Sn-based solder. The addition of Co-P/Ni-P layer successfully inhibited elemental diffusion between *p*-type Bi_2Te_3 and barrier layer, and prevented the erosion of the barrier layer by Sn from the solder. Additionally, *p*-type Bi_2Te_3 wafers with ECEN layer exhibited great wettability, facilitating the fabrication of thermoelectric devices. Furthermore, the ECEN-plated modules exhibited great bonding strength, ensuring high reliability of the joints at room temperature. Meanwhile, compared to Co-P or Ni-P single layer, no significant change in contact resistivity was identified for the joints of *p*-type Bi_2Te_3/Co-P/Ni-P/Sn95Sb5 solder. When annealed at 180 ℃ for 24 h, the contact resistivity of thermoelectric modules with a Co-P/Ni-P layer increases

slowly and shows excellent performance. These results confirm that Co-P/Ni-P can be an extremely promising and effective diffusion barrier for application in *p*-type Bi_2Te_3-based thermoelectric modules.

CRediT authorship contribution statement

Shui Jiachang: Visualization, Methodology, Formal analysis, Conceptualization, Writing – original draft, Validation, Investigation, Data curation. **Wei Wu:** Validation. **Xiaoming Hu:** Validation, Resources, Funding acquisition, Supervision, Investigation. **Fan Xi'an:** Resources, Funding acquisition, Supervision, Project administration. **Xin Chen:** Validation. **Zigui Luo:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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